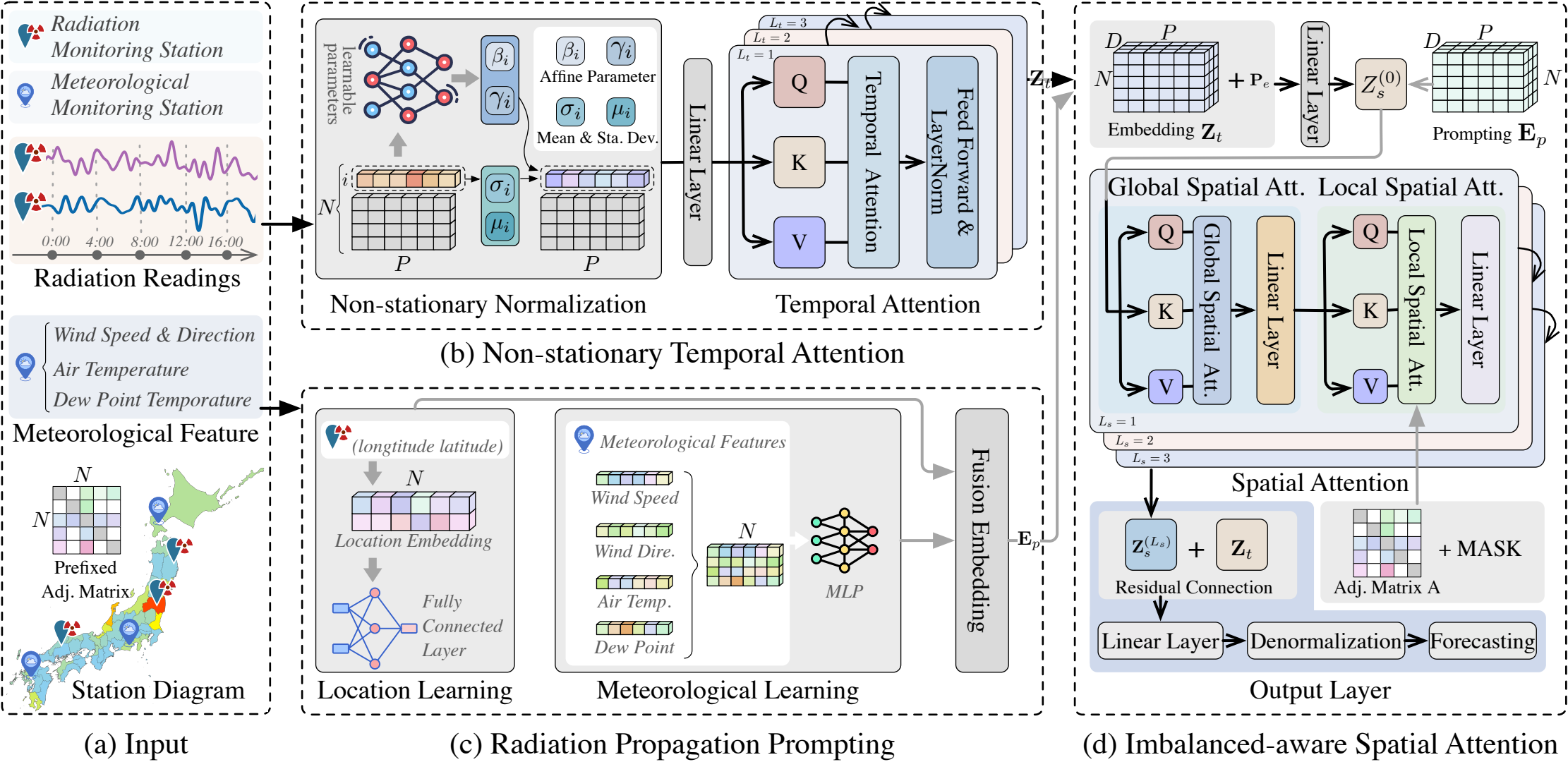
Radiation Monitoring Station
Meteorological Monitoring Station
0:00
4:00
8:00
12:00
16:00
Radiation Readings
Wind Speed & Direction
Air Temperature
Dew Point Temperature
Meteorological Feature
N
N
Prefixed Adj. Matrix
Station Diagram
learnable parameters
Affine Parameter
Mean & Sta. Dev.
P
P
Non-stationary Normalization
Linear Layer
Q
K
V
Temporal Attention
Feed Forward & LayerNorm
Temporal Attention
(b) Non-stationary Temporal Attention
(longtitude latitude)
Location Embedding
Fully Connected Layer
Location Learning
Meteorological Features
Wind Speed
Wind Dire.
Air Temp.
Dew Point
MLP
Meteorological Learning
Fusion Embedding
Embedding $\mathbf{Z}_t$
Prompting $\mathbf{E}_p$
Linear Layer
Global Spatial Att.
Local Spatial Att.
Spatial Attention
Residual Connection
Adj. Matrix A
+ MASK
Linear Layer
Denormalization
Forecasting
Output Layer
(a) Input
(c) Radiation Propagation Prompting
(d) Imbalanced-aware Spatial Attention